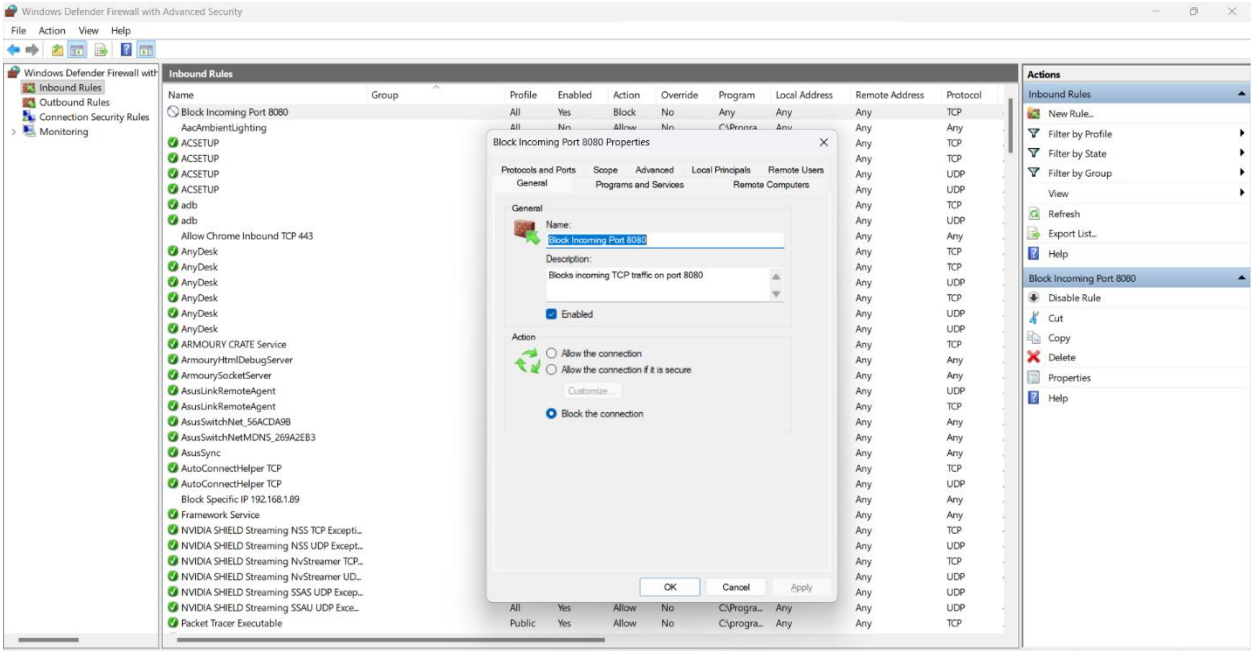
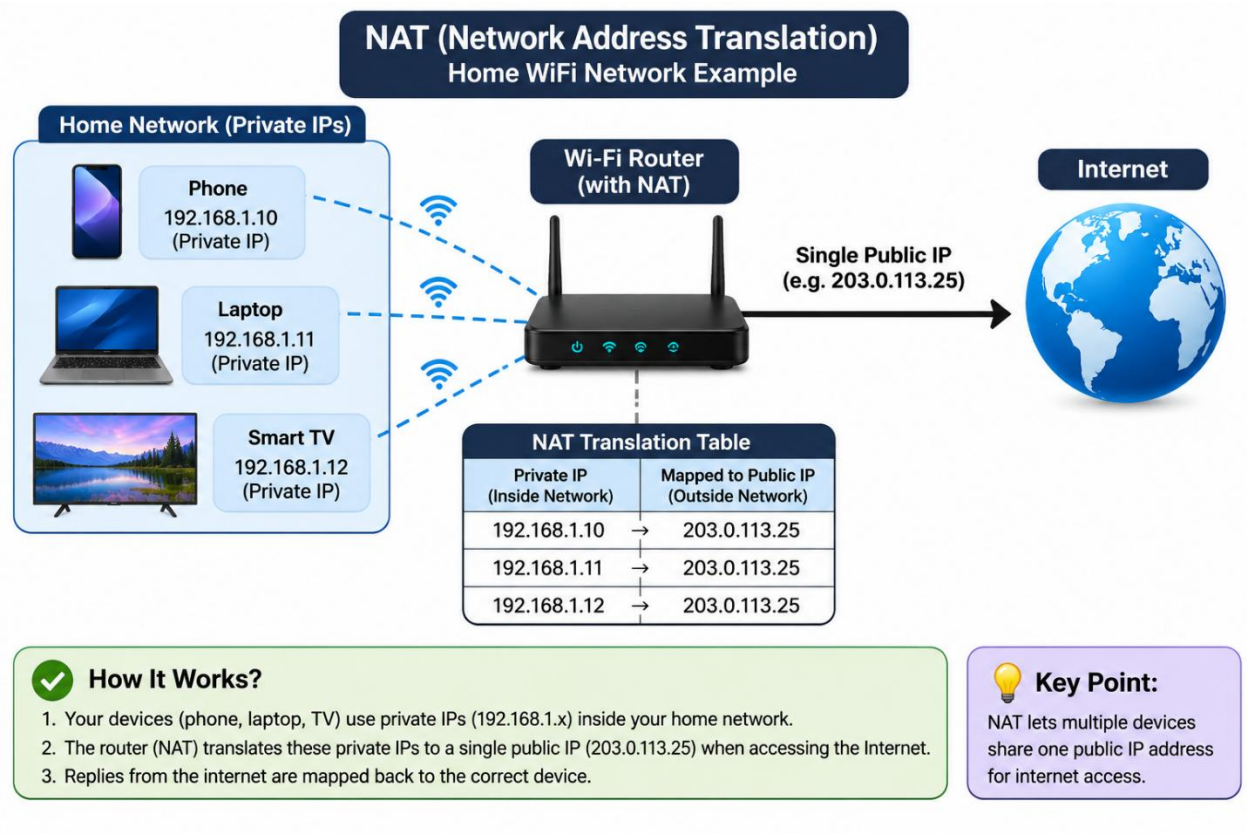
Secure networks with firewalls NAT, port filtering, packet filtering, and other methods

Task 1 : Set up Windows Defender Firewall on your Windows laptop or PC and create a new inbound rule to block all incoming traffic on port 8080. Take a screenshot of the rule you created.



Task 2 : Simulate NAT (Network Address Translation) by drawing a simple diagram showing how your home WiFi router assigns private IP addresses to your phone, laptop, and smart TV, and how it translates them to a single public IP for internet access.



Explanation of NAT

NAT (Network Address Translation) is a technique used by routers to translate private IP addresses into a public IP address.

- The Wi-Fi router assigns private IP addresses to connected devices.
- When a device accesses the internet, the router translates its private source IP address into the public IP address.
- Multiple devices can share a single public IP address using NAT, typically through PAT (Port Address Translation).

Example of NAT Translation

Device	Private IP Address	Public IP Address
Phone	192.168.1.2	49.36.100.25
Laptop	192.168.1.3	49.36.100.25
Smart TV	192.168.1.4	49.36.100.25

Note: These are fictional example IP addresses. All three devices share the same public IP address.

Advantages of NAT

1. Allows multiple devices to share one public IP address.
2. Helps conserve public IPv4 addresses.
3. Hides internal private IP addresses from the public internet.

Task 3 : Configure port filtering using your WiFi router's admin panel (or use a router simulator online if you don't have access). Block outgoing traffic on port 25 (SMTP) and explain in 2 sentences how this could help prevent spam from your network.

To configure port filtering on a Wi-Fi router to block outgoing traffic on port 25 (SMTP).

1. What is Port Filtering?

Port filtering is a network security technique used to allow or block network traffic based on specific port numbers.

Port 25 is commonly used by SMTP (Simple Mail Transfer Protocol) for sending emails.

2. Step-by-Step Configuration

Step 1: Access the Router Admin Panel

1. Connect the laptop to the Wi-Fi router.
2. Open a web browser.
3. Enter the router's IP address, such as 192.168.1.1.
4. Enter the administrator username and password.

Step 2: Open Security Settings

1. Open the Advanced Settings section.
2. Find Security, Access Control or Firewall settings.
3. Look for the Access Control or IP/Port Filtering option.

Step 3: Create a Port Filtering Rule

Setting	**Value**
Rule Name	Block SMTP Port 25
Direction	Outgoing
Protocol	TCP
Destination Port	25
Action	Deny / Block
Status	Enabled

Step 4: Save the Configuration

1. Click Add Rule or Save.
2. Enable the newly created rule.
3. Apply the settings.
4. Check whether the rule appears in the filtering rules list.

Note: Router menus differ by manufacturer and model. Some home routers do not support outgoing port filtering. If this feature is unavailable, a router simulator or another firewall supporting outbound rules can be used.

3. How Does Blocking Port 25 Prevent Spam?

Blocking outgoing traffic on port 25 can help prevent infected devices or malware from directly sending large amounts of spam emails through external mail servers. It reduces unauthorized SMTP connections from the network, although it does not block every method of sending spam.

4. Conclusion

Port filtering is an important network security technique. By blocking outgoing TCP traffic on port 25, network administrators can reduce unauthorized email activity and help protect the network from spam.

Task 4 : Given a sample packet filtering rule set (provided by your trainer), identify which of these three sample packets would be allowed or blocked: 1) HTTP request from 192.168.1.5 to 172.217.166.78:80, 2) SSH request from 10.0.0.10 to 34.201.23.1:22, 3) DNS query from 192.168.1.7 to 8.8.8.8:53.

To analyze sample network packets and identify whether they should be allowed or blocked based on packet filtering rules.

1. What is Packet Filtering?

Packet filtering is a network security technique used by firewalls to inspect incoming and outgoing data packets. It checks information such as source IP address, destination IP address, protocol and port number to decide whether to allow or block network traffic.

2. Sample Packet Filtering Rules

For this exercise, assume the following firewall rules:

Rule No.	Protocol	Destination Port	Action
1	TCP	80	Allow
2	TCP	22	Block
3	UDP	53	Allow
4	All Other Traffic	Any	Block

3. Packet Analysis

Packet 1: HTTP Request

- Source IP: 192.168.1.5
- Destination IP: 172.217.166.78
- Destination Port: 80
- Protocol: TCP

Decision: ALLOWED

Reason: Port 80 is used for HTTP communication, and the firewall rule allows TCP traffic on port 80.

Packet 2: SSH Request

- Source IP: 10.0.0.10
- Destination IP: 34.201.23.1
- Destination Port: 22
- Protocol: TCP

Decision: BLOCKED

Reason: Port 22 is used for SSH communication, and the firewall rule blocks TCP traffic on port 22.

Packet 3: DNS Query

- Source IP: 192.168.1.7
- Destination IP: 8.8.8.8
- Destination Port: 53
- Protocol: UDP

Decision: ALLOWED

Reason: Port 53 is used for DNS communication, and the firewall rule allows UDP traffic on port 53.

4. Final Result Table

Packet	Service	Destination Port	Decision
1	HTTP	80	Allowed
2	SSH	22	Blocked
3	DNS	53	Allowed

5. Conclusion

Packet filtering helps protect a network by controlling traffic based on IP addresses, protocols and port numbers. In this exercise, the HTTP and DNS packets are allowed, while the SSH packet is blocked according to the assumed firewall rules.

Task 5 : Research and list 3 ways Instagram or Zomato might use firewalls and packet filtering to protect their servers from attacks or misuse.

To understand how popular online platforms like Instagram and Zomato use firewalls and packet filtering to protect their servers from cyberattacks and misuse.

1. Blocking Unauthorized Network Traffic

Instagram and Zomato can use firewalls to block unauthorized incoming network connections. Firewalls allow only permitted traffic to reach their servers and help prevent unauthorized access.

2. Preventing DDoS Attacks

Both platforms can use firewalls and traffic filtering systems to detect and block suspicious traffic from multiple sources. This helps reduce Distributed Denial-of-Service (DDoS) attacks, which attempt to overload servers with excessive requests.

3. Filtering Suspicious Requests and Ports

Firewalls and packet filtering can be used to allow only required network protocols and ports while blocking unwanted traffic. This helps protect servers from unauthorized connections, suspicious requests and misuse of network services.

Conclusion

Firewalls and packet filtering are important parts of network security. Platforms like Instagram and Zomato can use these security methods to protect their servers, reduce cyberattacks, block unauthorized traffic and provide safer online services to users.